

Structural Features of Lithium Organoborates

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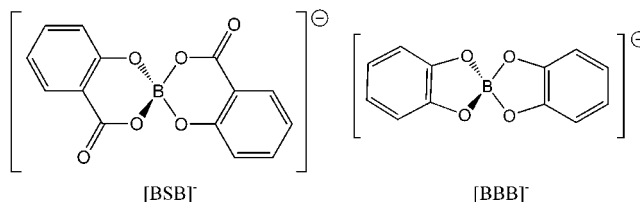
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ABSTRACT: The NMR and crystallographic characterization of lithium bis[salicylato(2-)]borate, Li(BSB), and lithium bis[1,2-benzenediolato(2-)-O,O']borate, Li(BBB), are reported. The single-strand polymer $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ crystallizes from a saturated solution of Li(BSB) in ethanenitrile and consists of $\mu_2\text{-}\{(\text{MeCN})_2\text{Li}\}^+$ cations bridging the exocyclic oxygen atoms of two $[\text{BSB}]^-$ anions. Crystallization of Li(BSB) from tetrahydrofuran produces $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$, which has essentially the same structure as $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$. In contrast, crystals of Li(BBB) grown from THF/diethyl ether or THF/toluene mixtures (1:1) contain adventitious water and form a hydrogen-bonded, three-dimensional lattice comprising $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$ units. This novel structural motif contains the first example of lithium bound to one THF and two water molecules, and also exhibits hydrogen bonding between the water and the THF. Ab initio calculations suggest that the Li–O_{BBB} bond in $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$, while still predominantly ionic in character, is more covalent than an analogous Na–O_{BBB} bond. Crystallization of Li(BBB) from ethanenitrile produces $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$, which contains the first example of lithium bound to one MeCN and one water molecule. In this case, hydrogen bonding occurs between dimeric $\{(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})\}_2$ units to form two-dimensional sheets in the *ab* crystal plane.

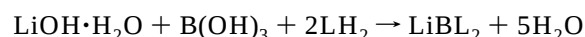
Introduction

Many of the materials employed in traditional lithium battery electrolytes suffer from serious practical drawbacks comprising safety risks, environmental concerns, stability problems, and cost.^{1,2} Examples of such compounds investigation include $\text{Li}[\text{ClO}_4]$, $\text{Li}[\text{AsF}_6]$, $\text{Li}[\text{PF}_6]$, $\text{Li}[\text{BF}_4]$, and more recently, “triflate”, $\text{Li}[(\text{CF}_3\text{SO}_3)]$, “bistriflamide”, $\text{Li}[\text{N}(\text{SO}_2\text{CF}_3)_2]$, and “methide”, $\text{Li}[\text{C}(\text{SO}_2\text{CF}_3)_3]$,^{1,2} among others. While many of these systems are still being actively investigated, the search for suitable substitutes remains a topic of considerable industrial and academic significance. In a series of papers and patents published over the past few years, a range of lithium organoborates has been considered for this purpose.^{3,4} The anions of interest are bis(chelate) complexes of boron based on aromatic or aliphatic diols, carboxylic acids, and related ligands. Such materials are usually nontoxic, inexpensive, and thermally, chemically, and electrochemically stable. Examples include lithium bis[salicylato(2-)]borate, Li(BSB), and lithium bis[1,2-benzenediolato(2-)-O,O']borate, Li(BBB)^{3,4} (below). While knowledge of the structures of these compounds is clearly a useful precursor to understanding their behavior, previous studies have focused mainly on their electrochemistry,^{3,4} and structural characterization has been limited. Here the solid state behavior of Li(BSB) and Li(BBB) in response to solvation by ethanenitrile (MeCN) and tetrahydrofuran (THF) is investigated.

A range of lithium organoborates can be prepared, using the methods reported by Wühr³ and Gores et



al.,^{4a,4d} from lithium hydroxide, boric acid, and the protonated form of the desired ligand, LH_2 [for Li(BSB), LH_2 = salicylic acid; for Li(BBB), LH_2 = catechol]:



Although a survey of the Cambridge Crystallography Database⁵ revealed seven occurrences of the $[\text{BSB}]^-$ anion⁶ and fifteen of the $[\text{BBB}]^-$ anion,⁷ no crystal structures have been reported previously for the lithium salts Li(BSB) and Li(BBB). In addition to their relevance to lithium batteries, the solid-state architectures of lithium organoborates are also of interest from the perspective of organolithium chemistry, particularly with reference to laddering and solvation effects.⁸ Here we report crystallographic characterization of the polymeric solvates $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ and $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$, as well as the unusual mixed solvates $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$ and $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$. The ¹H, ⁷Li, ¹¹B, and ¹³C NMR spectral data for Li(BSB) and Li(BBB) are also presented.

Experimental Section

Lithium Bis[salicylato(2-)]borate, Li(BSB). A sample of Li(BSB) was prepared from lithium hydroxide monohydrate, salicylic acid, and boric acid according to published pro-

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Table 1. Crystal Data and Structure Refinement for $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$, $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$, $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$, and $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$

	$\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$	$\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$	$\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$	$\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$
empirical formula	$\text{C}_{18}\text{H}_{14}\text{BLiN}_4\text{O}_6$	$\text{C}_{22}\text{H}_{24}\text{BLiO}_8$	$\text{C}_{16}\text{H}_{20}\text{BLiO}_7$	$\text{C}_{15}\text{H}_{10}\text{BLiNO}_4$
molecular weight/g mol ⁻¹	400.08	434.16	342.07	285.99
crystal system	triclinic	monoclinic	triclinic	triclinic
space group	<i>P</i> -1	<i>Cc</i>	<i>P</i> -1	<i>P</i> -1
<i>a</i> /Å	8.8049(14)	8.847(2)	7.9406(19)	7.8764(8)
<i>b</i> /Å	9.9783(16)	17.329(6)	9.919(3)	7.9089(8)
<i>c</i> /Å	11.8560(19)	14.646(4)	11.961(3)	12.3566(13)
α /°	105.777(3)	90	79.233(4)	106.212(2)
β /°	92.197(3)	92.775(4)	70.754(7)	90.475(2)
γ /°	112.402(3)	90	71.591(5)	94.864(2)
volume/Å ³	914.9(3)	2249.6(11)	840.3(4)	736.06(13)
<i>Z</i>	2	4	2	2
<i>D</i> _{calc} /g cm ⁻³	1.452	1.282	1.352	1.290
absorption coefficient/mm ⁻¹	0.109	0.095	0.103	0.092
<i>F</i> ₀₀₀	412	912	360	294
crystal size/mm	0.45 × 0.32 × 0.24	0.26 × 0.23 × 0.18	0.62 × 0.48 × 0.22	0.45 × 0.32 × 0.24
θ range/°	1.81 to 28.67	2.35 to 28.21	1.81 to 28.37	1.72 to 28.68
reflections collected	7184	6916	9165	8683
independent reflections	3954	4346	3668	3334
<i>R</i> (int)	0.0685	0.0915	0.0459	0.0608
completeness to θ	84.0	89.6	87.3	87.7
absorption correction	none	none	none	none
data/restraints/parameters	3954/0/309	4346/2/289	3668/0/306	3334/0/251
goodness-of-fit on <i>F</i> ²	1.616	1.037	1.044	0.865
final <i>R</i> indices [<i>I</i> > 2 σ (<i>I</i>)] ^a	<i>R</i> ₁ = 0.0654 <i>wR</i> ₂ ^b = 0.2425	<i>R</i> ₁ = 0.0619 <i>wR</i> ₂ ^c = 0.1525	<i>R</i> ₁ = 0.0534 <i>wR</i> ₂ ^d = 0.1381	<i>R</i> ₁ = 0.0525 <i>wR</i> ₂ ^b = 0.1291
<i>R</i> indices (all data) ^a	<i>R</i> ₁ = 0.0838 <i>wR</i> ₂ ^b = 0.2500	<i>R</i> ₁ = 0.0835 <i>wR</i> ₂ ^c = 0.1735	<i>R</i> ₁ = 0.0666 <i>wR</i> ₂ ^d = 0.1478	<i>R</i> ₁ = 0.0953 <i>wR</i> ₂ ^b = 0.1576
largest diff. peak and hole/eÅ ⁻³	0.364 and -0.333	0.363 and -0.267	0.410 and -0.322	0.308 and -0.280

^a $R = \sum ||F_o| - |F_c|| / \sum |F_o|$ and $wR = [\sum (|F_o| - |F_c|)^2 / \sum w F_o^2]^{1/2}$. ^b $w = [\sigma^2(F_o^2) + (0.1000P)^2]^{-1}$ where $P = (F_o^2 + 2F_c^2)/3$. ^c $w = [\sigma^2(F_o^2) + (0.0969P)^2]^{-1}$ where $P = (F_o^2 + 2F_c^2)/3$. ^d $w = [\sigma^2(F_o^2) + (0.0996P)^2]^{-1}$ where $P = (F_o^2 + 2F_c^2)/3$.

cedures,^{3,4a,4d} and used without further purification. ¹H NMR [300 MHz, 25 °C, δ (TMS in CDCl₃)] in D₂O: δ /ppm = 7.89–7.62 (m, 2H), 7.40–7.35 (m, 2H), 6.83–6.70 (m, 4H). ¹³C NMR [75.47 MHz, 25 °C, δ (TMS in CDCl₃)] in DMSO-*d*₆: δ /ppm = 164.33 (s, C=O), 159.31 (s), 135.17 (s), 129.63 (s), 119.28 (s), 118.41 (s), 115.58 (s). ⁷Li NMR [194.37 MHz, 25 °C, δ (1 M LiCl in D₂O)] in CD₃CN: δ /ppm = -1.79. ¹¹B-NMR [160.46 MHz, 25 °C, δ (BF₃O(C₂H₅)₂ in D₂O)] in CD₃CN: δ /ppm = 0.05.

The white powder Li(BSB) dissolved in water, propanone, propan-2-ol, THF, and ethanenitrile; it was insoluble in diethyl ether, dichloromethane, trichloromethane, hexane, and toluene. Hydrolysis was observed on standing in water, as verified by ¹H and ¹³C NMR spectroscopy, and by comparing an observed unit cell of the salicylic acid produced to the published X-ray crystal structure.⁹ Crystals of $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ suitable for X-ray diffraction experiments were deposited from a saturated solution of Li(BSB) in ethanenitrile stored at 4 °C for several days. Crystals of $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$ were prepared similarly from a saturated solution of Li(BSB) in THF.

Lithium Bis[1,2-benzenediolato(2-)-O,O']borate, Li(BBB). A sample of Li(BBB) was prepared from lithium hydroxide monohydrate, catechol, and boric acid according to published procedures,^{3,4a,4d} and used without further purification. ¹H NMR [300 MHz, 25 °C, δ (TMS in CDCl₃)] in CD₃CN: δ /ppm = 6.60 (s). ¹³C NMR [75.47 MHz, 25 °C, δ (TMS in CDCl₃)] in CD₃CN: δ /ppm = 151.02 (s), 117.01 (s), 107.73 (s). ⁷Li NMR [194.37 MHz, 25 °C, δ (1 M LiCl in D₂O)] in CD₃CN: δ /ppm = -2.36. ¹¹B NMR [160.46 MHz, 25 °C, δ (BF₃O(C₂H₅)₂ in D₂O)] in CD₃CN: δ /ppm = 10.42.

The white powder Li(BBB) dissolved in water, propan-2-ol, diethyl ether, THF, and ethanenitrile; it was insoluble in dichloromethane, trichloromethane, hexane, and toluene. Hydrolysis to catechol was observed on standing in water (verified by ¹H and ¹³C NMR spectroscopy), and dissolution in propanone gave a clear, blue-green solution with evolution of gas. Crystals of $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$ suitable for X-ray diffraction experiments were deposited from a saturated solution of Li(BBB) in THF/diethyl ether or THF/toluene (both 1:1) stored at 4 °C for several days. Crystals of $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$ were grown from a saturated solution of Li(BBB) in ethane-

nitrile stored at -10 °C for periods in excess of one month. In all cases, the crystals redissolved quickly on warming to room temperature and became opaque when removed from their mother liquor.

X-ray Measurements. Colorless crystals of $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$, $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$, $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$, and $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$ were coated in oil (Paratone 8277, Exxon) and mounted on glass fibers. Data were collected at low temperature (-120 °C) using a Bruker AXS Smart CCD diffractometer with graphite monochromated Mo K α radiation (λ = 0.71073 Å). The structures were solved by direct methods and expanded using full-matrix least-squares on *F*².¹⁰ Further details of the crystal data sets and structure refinements are given in Table 1, and pertinent bond lengths and angles are summarized in Table 2. The hydrogen atoms of the water molecules in $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$ and $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$ were all located and refined; relevant data are summarized in Table 3. All other hydrogen atoms were included at geometrically idealized positions and were not refined.

Calculations. The ab initio package GAMESS¹¹ was used to determine the location of electron density, and MOLEKEL¹² was used as a visualization tool. The basis set used was Huzinaga's three Gaussian expansion set,¹³ since the exponents and contractions are optimized separately to yield lower energies than the usual STO-3G. To achieve higher accuracy in the Mulliken bonding orders and atomic charges,¹⁴ Møller-Plesset second-order perturbation theory was employed. All calculations were performed in Cartesian coordinates on single monomer units.

Results and Discussion

The lithium organoborates Li(BSB) and Li(BBB) were prepared from lithium hydroxide monohydrate, boric acid, and salicylic acid or catechol, respectively, according to previously published procedures.^{3,4a,4d} Both salts were obtained as white powders in essentially quantitative yield and used without further purification. The ¹H,

Table 2. Selected Bond Lengths (in Å) and Angles (in °) for $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$, $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$, $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$, and $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$

$\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty^a$		$\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty^b$		$\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$		$\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty^c$	
B(1)–O(1A)	1.439(4)	B(1)–O(9A)	1.459(5)	B(1)–O(1A)	1.467(2)	B(1)–O(1A)	1.476(3)
B(1)–O(1B)	1.447(4)	B(1)–O(9B)	1.472(5)	B(1)–O(1B)	1.4674(19)	B(1)–O(1B)	1.478(3)
B(1)–O(81B)	1.488(4)	B(1)–O(1B)	1.491(4)	B(1)–O(8B)	1.468(2)	B(1)–O(8B)	1.476(3)
B(1)–O(81A)	1.492(4)	B(1)–O(1A)	1.494(5)	B(1)–O(8A)	1.4685(19)	B(1)–O(8A)	1.486(3)
C(2A)–O(1A)	1.351(3)	C(2A)–O(1A)	1.337(4)	C(2A)–O(1A)	1.3605(17)	C(2A)–O(1A)	1.361(2)
C(2B)–O(1B)	1.355(3)	C(2B)–O(1B)	1.337(4)	C(7A)–O(8A)	1.3508(18)	C(7A)–O(8A)	1.370(2)
C(8A)–O(82A)	1.220(3)	C(2A)–O(2A)	1.240(4)	C(2B)–O(1B)	1.3556(18)	C(2B)–O(1B)	1.382(2)
C(8B)–O(82B)	1.226(3)	C(2B)–O(2B)	1.250(4)	C(7B)–O(8B)	1.3492(18)	C(7B)–O(8B)	1.367(2)
C(8A)–O(81A)	1.315(3)	C(8A)–O(9A)	1.359(4)				
C(8B)–O(81B)	1.315(3)	C(8B)–O(9B)	1.356(4)				
Li(1)#1–O(82A)	1.894(5)	Li(1)–O(2A)	1.919(7)	Li(1)–O(1A)	1.923(3)	Li(1)–O(1B)	1.997(4)
Li(1)–O(82B)	1.866(5)	Li(1)–O(2B)#1	1.935(7)	Li(1)–O(1W)	1.863(3)	Li(1)–O(8A)#1	1.977(4)
Li(1)–N(4S)	2.057(6)	Li(1)–O(1S)	1.929(7)	Li(1)–O(2W)	1.888(3)	Li(1)–O(1W)	1.882(4)
Li(1)–N(1S)	2.075(6)	Li(1)–O(6S)	1.948(7)	Li(1)–O(1S)	1.932(3)	Li(1)–N(1S)	1.997(4)
O(1B)–B(1)–O(1A)	108.1(2)	O(9A)–B(1)–O(9B)	110.8(3)	O(1A)–B(1)–O(1B)	112.39(12)	O(1A)–B(1)–O(1B)	112.94(17)
O(1B)–B(1)–O(81B)	114.1(2)	O(9A)–B(1)–O(1B)	107.0(3)	O(1A)–B(1)–O(8B)	112.28(12)	O(8B)–B(1)–O(1A)	111.02(17)
O(1A)–B(1)–O(81B)	108.0(2)	O(9B)–B(1)–O(1B)	113.2(3)	O(1B)–B(1)–O(8B)	104.48(12)	O(8B)–B(1)–O(1B)	104.59(17)
O(1B)–B(1)–O(81A)	108.0(2)	O(9A)–B(1)–O(1A)	113.3(3)	O(1A)–B(1)–O(8A)	104.47(12)	O(1A)–B(1)–O(8A)	104.43(17)
O(1A)–B(1)–O(81A)	111.8(2)	O(9B)–B(1)–O(1A)	107.7(3)	O(1B)–B(1)–O(8A)	111.45(12)	O(1B)–B(1)–O(8A)	112.30(17)
O(81B)–B(1)–O(81A)	106.9(2)	O(1B)–B(1)–O(1A)	104.9(3)	O(8B)–B(1)–O(8A)	111.96(12)	O(8B)–B(1)–O(8A)	111.74(17)
O(82B)–Li(1)–O(82A)#2	110.2(3)	O(2A)–Li(1)–O(1S)	108.7(3)	O(1W)–Li(1)–O(2W)	114.82(14)	O(1W)–Li(1)–O(8A)#1	110.29(19)
O(82B)–Li(1)–N(4S)	106.7(3)	O(2A)–Li(1)–O(2B)#1	116.5(3)	O(1W)–Li(1)–O(1A)	111.65(13)	O(1W)–Li(1)–O(1B)	105.17(17)
O(82A)#2–Li(1)–N(4S)	111.3(3)	O(1S)–Li(1)–O(2B)#1	103.8(3)	O(2W)–Li(1)–O(1A)	105.92(13)	O(8A)#1–Li(1)–O(1B)	113.96(18)
O(82B)–Li(1)–N(1S)	112.5(3)	O(2A)–Li(1)–O(6S)	110.1(3)	O(1W)–Li(1)–O(1S)	107.74(13)	O(1W)–Li(1)–N(1S)	113.58(19)
O(82A)#2–Li(1)–N(1S)	98.9(2)	O(1S)–Li(1)–O(6S)	110.7(3)	O(2W)–Li(1)–O(1S)	102.33(12)	O(8A)#1–Li(1)–N(1S)	106.57(17)
N(4S)–Li(1)–N(1S)	117.1(3)	O(2B)#1–Li(1)–O(6S)	106.8(3)	O(1A)–Li(1)–O(1S)	114.20(13)	O(1B)–Li(1)–N(1S)	107.40(18)

^a Symmetry transformations used to generate equivalent atoms: #1, $x + 1, y, z$; #2, $x - 1, y, z$. ^b Symmetry transformations used to generate equivalent atoms: #1, $x + 1/2, y + 1/2, z$; #2, $x - 1/2, y - 1/2, z$. ^c Symmetry transformations used to generate equivalent atoms: #1, $-x, -y + 1, -z$.

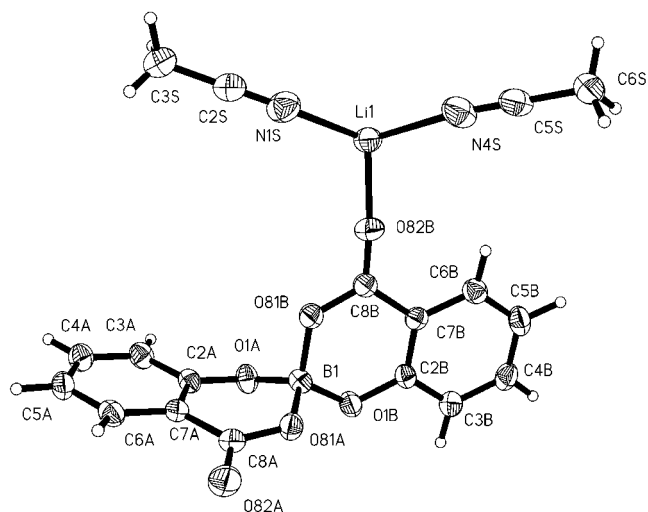
Table 3. Hydrogen Bonding Parameters in $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$ and $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$

$r(\text{O} \cdots \text{H})$ [Å]	$r(\text{H} \cdots \text{O})$ [Å]	$\angle(\text{O} \cdots \text{H} \cdots \text{O})$ [°]	$r(\text{O}/\text{O})$ [Å]	symmetry transformations		
{(H ₂ O) ₂ (THF)Li(BBB)} _∞						
O(1W)–H(1W1)	0.856	H(1W1)⋯O8A	1.984	161.56	2.809	$x + 1, y, z$
O(1W)–H(1W2)	0.858	H(1W2)⋯O8B	1.933	167.41	2.777	$-x + 1, -y + 1, -z$
O(2W)–H(2W1)	0.878	H(2W1)⋯O1B	1.822	171.43	2.693	$-x + 1, -y, -z$
O(2W)–H(2W2)	0.877	H(2W2)⋯O1S	1.959	175.89	2.835	$-x + 2, -y, -z$
{[(H ₂ O)(MeCN)Li(BBB)] ₂ } _∞						
O(1W)–H(1W1)	0.901	H(1W1)⋯O(1A)	1.905	173.61	2.802	$-x + 1, -y + 1, -z$
O(1W)–H(1W2)	0.942	H(1W2)⋯O(8B)	1.871	168.47	2.800	$x, y - 1, z$

^7Li , ^{11}B , and ^{13}C NMR spectra for $\text{Li}(\text{BSB})$ and $\text{Li}(\text{BBB})$ were unremarkable, comprising a single set of resonances in each case. The ^1H , ^7Li , and ^{13}C NMR signals for both compounds fall in predicted regions, and the peaks at 0.05 ppm (former) and 10.42 ppm (latter) in the ^{11}B NMR spectra are typical for complexes containing four-coordinate boron.¹⁵

To better understand the behavior of these compounds, we qualitatively examined their solubility in a variety of solvents and grew crystals suitable for single-crystal X-ray diffraction studies. Figure 1 shows the adduct $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$, obtained on recrystallization of $\text{Li}(\text{BSB})$ from ethanenitrile. The μ_2 - $\{(\text{MeCN})_2\text{Li}\}^+$ cation in $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ is coordinated to two solvent molecules [$\bar{r}(\text{Li} \cdots \text{N}) = 2.066(6)$ Å], and bridges the exocyclic oxygen atoms of two $[\text{BSB}]^-$ anions [$\bar{r}(\text{Li} \cdots \text{O}) = 1.880(5)$ Å]. Each $[\text{BSB}]^-$ itself bridges two lithium centers, thus forming a linear polymer (Figure 2). When $\text{Li}(\text{BSB})$ is recrystallized from THF, the resulting coordination polymer $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$ contains a μ_2 - $\{(\text{THF})_2\text{Li}\}^+$ cation, but otherwise adopts approximately the same structure as $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ (Figures 3 and 4).

While solvation of lithium cations by THF is well-known (the Cambridge Crystallography Database contains over 6000 cases⁵), coordination by MeCN is relatively rare. Of 10 previous examples,¹⁶ only the bis-

**Figure 1.** Thermal ellipsoid plot (50% probability) of one monomer unit contained in the crystal structure of $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$, showing the atom labeling scheme.

(ethanenitrile) adducts of lithium chloride $\{(\text{MeCN})_2\text{LiCl}\}_\infty$ ¹⁷ and lithium dicyanamide $\{(\text{MeCN})_2\text{Li}[\text{N}(\text{CN})_2]\}_\infty$ ¹⁸ are polymeric. Despite obviously different geometric profiles, the anions in $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$, $\{(\text{MeCN})_2\text{LiCl}\}_\infty$, and $\{(\text{MeCN})_2\text{Li}[\text{N}(\text{CN})_2]\}_\infty$ all act as

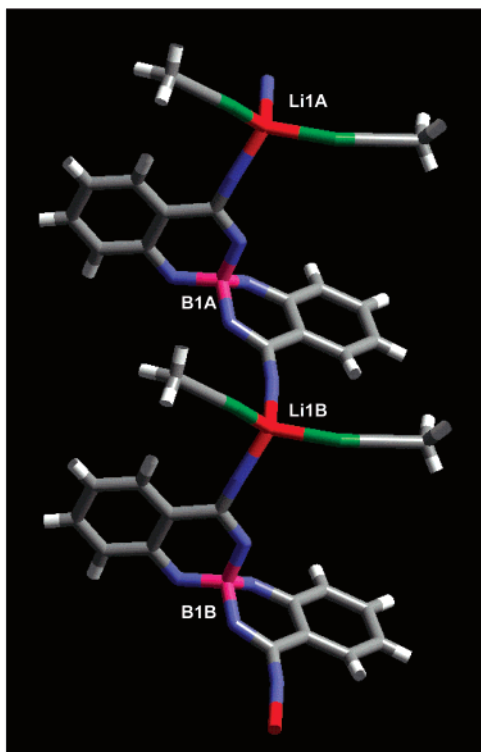


Figure 2. View of two units of $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ looking down the c -axis, orthogonal to the vector of the polymer chain (a -axis). Color scheme: white, hydrogen; red, lithium; pink, boron; gray, carbon; blue, oxygen; green, nitrogen.

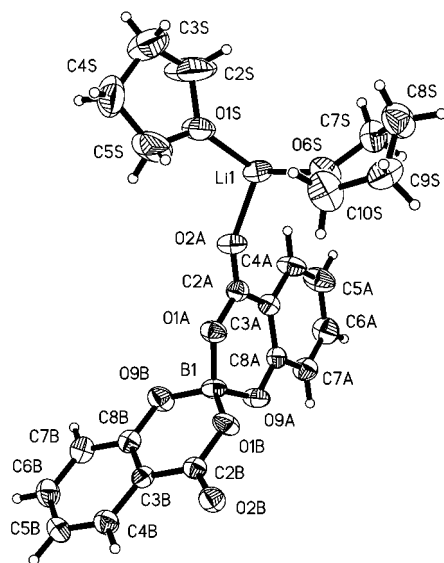
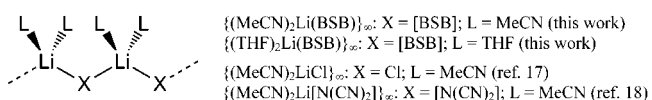


Figure 3. Thermal ellipsoid plot (50% probability) of one monomer unit contained in the crystal structure of $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$, showing the atom labeling scheme.

bridges between $\mu_2\text{-}\{(\text{MeCN})_2\text{Li}\}^+$ cations in essentially the same manner (below).



The arrangements adopted by all three bis(ethanenitrile) polymeric adducts are similar. For example, although the bridging anion in $\{(\text{MeCN})_2\text{LiCl}\}_\infty$ is

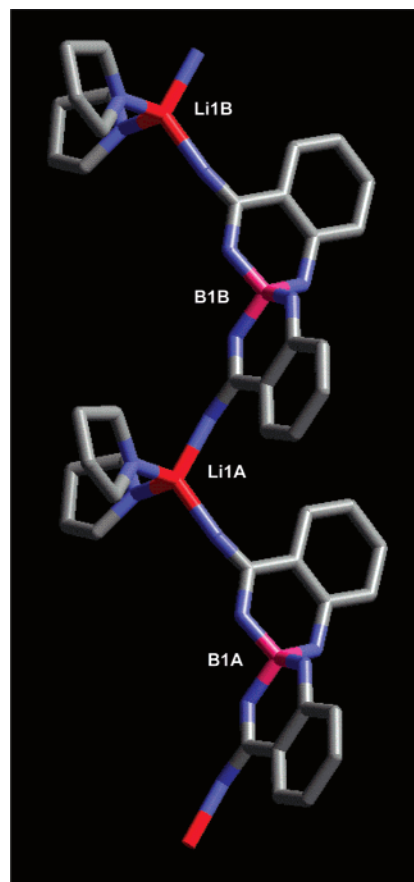


Figure 4. View of two units of $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$ looking down the a -axis, orthogonal to the vector of the polymer chain (b -axis). Hydrogen atoms have been omitted for clarity. Color scheme as in Figure 2.

μ_2 -chloride rather than an organoborate, the chain angle at lithium [$\angle(\text{Cl}-\text{Li}-\text{Cl}) = 111.1(2)^\circ$] is still very close to the corresponding angle in $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ [$\angle(\text{O}-\text{Li}-\text{O}) = 110.2(3)^\circ$]. Likewise, the dicyanamide anions in $\{(\text{MeCN})_2\text{Li}[\text{N}(\text{CN})_2]\}_\infty$ bridge $\mu_2\text{-}\{(\text{MeCN})_2\text{Li}\}^+$ cations through the two cyano nitrogen atoms of $[\text{N}(\text{CN})_2]^-$, much as the [BSB][−] anion utilizes two exocyclic oxygen atoms in $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$. The same basic architecture is in fact known for a variety of other solvents and anions as well, e.g., $\{(\text{C}_5\text{H}_5\text{N})_2\text{Li}[\text{CN}]\}_\infty$,¹⁸ the lithium thioamidate $\{(\text{THF})_2\text{Li}[\text{SC}(\text{Me})(\text{NMe})]\}_\infty$,¹⁹ and the lithium thiolate $\{(\text{NC}_5\text{H}_5)_2\text{Li}[\text{SPh}]\}_\infty$.²⁰

The fact that a similar polymeric structure is retained for $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ when THF is substituted for MeCN is probably a consequence of the steric profile of the [BSB][−] ligand. In contrast, when MeCN is replaced by THF in $\{(\text{MeCN})_2\text{LiCl}\}_\infty$, a discrete dimer $\{(\text{THF})_2\text{LiCl}\}_2$ comprising a four-membered $\{\text{Li}_2\text{Cl}_2\}$ ring forms.²¹ While this is thought to be a consequence of the increased steric demand of THF relative to MeCN in $\{(\text{MeCN})_2\text{LiCl}\}_\infty$,^{17a} the lithium centers in $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ and $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$ are apparently far enough apart that the steric abutment of solvent molecules is inconsequential. The larger, more tightly coordinated solvent molecules in $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$ [$\bar{r}(\text{Li}-\text{O}_{\text{THF}}) = 1.938(7)$ Å] do, however, result in a slight lengthening of the Li–O bonds between $\mu_2\text{-}\{(\text{THF})_2\text{Li}\}^+$ cations and [BSB][−] anions [$\bar{r}(\text{Li}-\text{O}_{\text{BSB}}) = 1.927(7)$ Å] and an increase

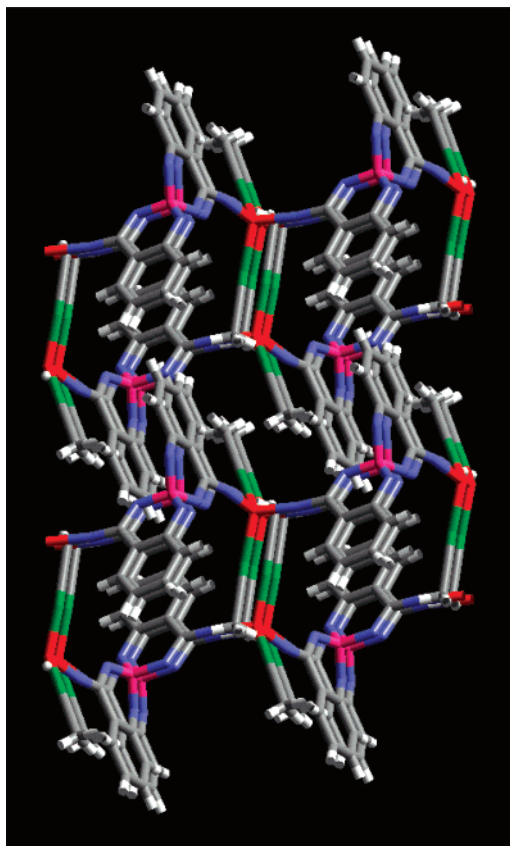


Figure 5. Packing in the crystal structure of $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$, looking down the a -axis, showing inter-ring interactions of $[\text{BSB}]^-$ and organization of solvent into channels. Color scheme as in Figure 2.

in the chain angle at lithium [$\angle(\text{O}-\text{Li}-\text{O}) = 116.5(3)^\circ$] relative to $\{(\text{MeCN})_2\text{LiBSB}\}_\infty$.

Solvation also exerts an influence on the structures of $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ and $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$ by virtue of increased steric congestion around lithium. The unsolvated cesium salt of $[\text{BSB}]^-$ also contains cations bound to the exocyclic oxygen atoms of the anion, but the $\text{Cs}-\text{O}$ bonds thus formed interact weakly to form $\{\text{Cs}_2\text{O}_2\}$ rings [$r(\text{Cs}-\text{O}) = 3.058 \text{ \AA}$, $r(\text{Cs}\cdots\text{O}) = 3.763 \text{ \AA}$].^{6g} Such lateral association, or "laddering", of $\text{Li}-\text{O}$ bonds in $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ and $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$ is prevented by the protrusion of solvent from the polymer backbone. The extended packing arrangements are instead dominated by the organization of solvent into channels and inter-ring interactions between anions (Figure 5). The cation in potassium bis(*p*-aminosalicylato)borate monohydrate is also solvated, coordinating only very weakly to the exocyclic oxygen of the anion [$r(\text{K}\cdots\text{O}) = 3.06 \text{ \AA}$], but rather more strongly to both a water molecule [$r(\text{K}-\text{O}) = 2.82 \text{ \AA}$] and a phenyl group in an η^6 -fashion.^{6b}

Bond distances within the $[\text{BSB}]^-$ units of $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ and $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$ are similar [e.g., $\bar{r}(\text{B}-\text{O}) = 1.466(4) \text{ \AA}$, $1.479(5) \text{ \AA}$, respectively] and correspond well with the mean distance observed for other $[\text{BSB}]^-$ complexes [$\bar{r}(\text{B}-\text{O}) = 1.46(2) \text{ \AA}$].⁶ Likewise, the exocyclic $\text{C}-\text{O}$ and contiguous endocyclic $\text{C}-\text{O}$ bond lengths [$\bar{r}(\text{C}-\text{O}) = 1.234(4)$, $1.336(4) \text{ \AA}$, respectively] are similar to the pertinent mean values [$\bar{r}(\text{C}-\text{O}) = 1.22(1)$, $1.32(2) \text{ \AA}$].⁶ Indeed, these distances do not vary significantly even in extreme cases such as $[\text{Zn}(\text{H}_2\text{O})_6][\text{BSB}]_2 \cdot 4\text{H}_2\text{O}$,

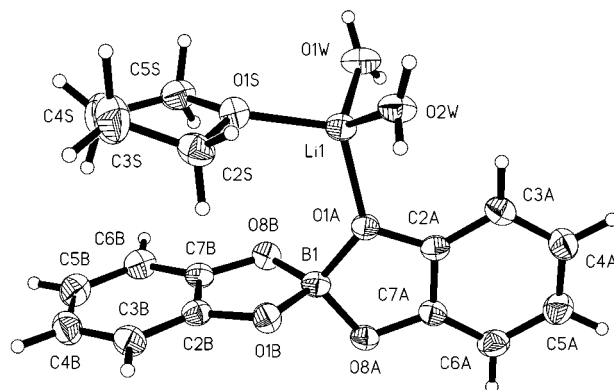


Figure 6. Thermal ellipsoid plot (50% probability) of one monomer unit contained in the crystal structure of $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$, showing the atom labeling scheme.

where $[\text{BSB}]^-$ is a discrete, noncoordinating anion [$\bar{r}(\text{C}-\text{O}) = 1.230$, $1.291 \text{ \AA}$].^{6d}

Unlike $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$ and $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$, the unusual mixed solvates $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BSB})\}$ and $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$ derive their stability from hydrogen bonding. Both $\text{Li}(\text{BBB})$ complexes incorporate adventitious water molecules when crystallized from wet solvents, and use these to assemble extended lattices through $\text{O}_{\text{water}}-\text{H}\cdots\text{O}_{\text{BBB}}$ contacts. As this phenomenon does not occur for $\text{Li}(\text{BSB})$ crystallized under the same conditions, it is most likely a result of demands imposed by the $[\text{BBB}]^-$ anion, which has four equivalent oxygen atoms. Water is employed to achieve coordinative saturation of both the anion (through hydrogen bonding) and cation (through solvation of lithium) in $\text{Li}(\text{BBB})$, which would otherwise be unattainable. The number of water molecules required for this task, and the nature of the resulting aggregate, depends on the cosolvent employed.

In the case of $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$, each lithium atom is bound to one oxygen atom of $[\text{BBB}]^-$ [$r(\text{Li}-\text{O}_{\text{BBB}}) = 1.923(3) \text{ \AA}$], one molecule of THF [$r(\text{Li}-\text{O}_{\text{THF}}) = 1.932(3) \text{ \AA}$], and two molecules of water [$\bar{r}(\text{Li}-\text{O}_{\text{water}}) = 1.875(3) \text{ \AA}$] (Figure 6). One of these water molecules is in turn hydrogen-bonded to oxygen atoms in two other $[\text{BBB}]^-$ anions [$r(\text{H}\cdots\text{O}_{\text{BBB}}) = 1.933$, $1.984 \text{ \AA}$], while the other is hydrogen bonded to one $[\text{BBB}]^-$ anion [$r(\text{H}\cdots\text{O}_{\text{BBB}}) = 1.822 \text{ \AA}$] and one molecule of THF [$r(\text{H}\cdots\text{O}_{\text{THF}}) = 1.959 \text{ \AA}$]. In this way, each $[\text{BBB}]^-$ anion is associated through its oxygen atoms with three water molecules and one lithium cation to form a three-dimensional lattice (Figure 7). This contrasts sharply with the situation observed for $\text{Li}(\text{BSB})$, in which no hydrogen bonding occurs in the lattice, even when wet solvents are used for crystallization (Figure 5).

The role of the THF molecules in $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$ is extremely unusual. We are surprised to report that the Cambridge Crystallography Database⁵ records no previous examples of hydrogen bonding involving coordinated THF, although a few examples of the "tetrahydrofuranium" cation $[\text{C}_4\text{H}_8\text{OH}]^+$ are known.²² In the present case, the THF molecules complete the coordination sphere of the four-coordinate lithium atoms and occupy the space between two $[\text{BBB}]^-$ phenyl rings. Each $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}$ unit acts as a hydrogen-bond donor to four other such units via water molecules, and a hydrogen bond acceptor via three of the oxygen

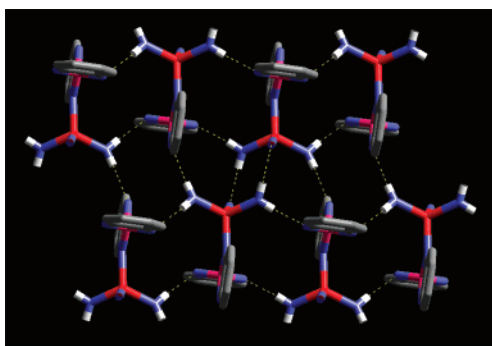


Figure 7. A portion of the of the hydrogen-bonded packing arrangement observed in the crystal structure of $\{(H_2O)_2(THF)Li(BBB)\}_\infty$ looking down the c -axis. The carbon atoms of THF and all hydrogen atoms except those on water have been omitted for clarity. Color scheme as in Figure 2.

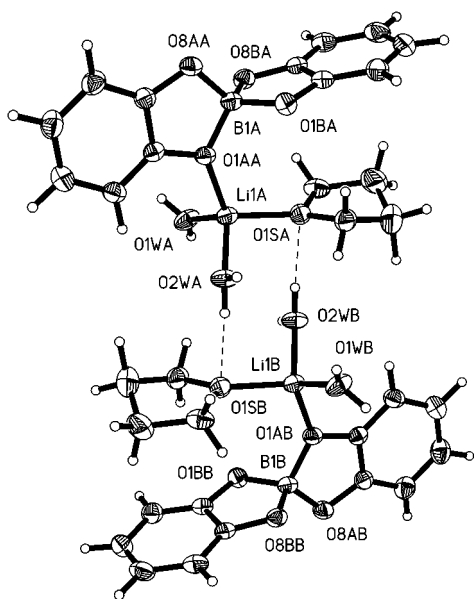


Figure 8. Thermal ellipsoid plot (50% probability) showing the hydrogen bonded dimer formed by two $\{(H_2O)_2(THF)Li(BBB)\}$ units. The dashed lines show hydrogen bonding between THF and water.

atoms of the $[BBB]^-$ anion and the oxygen atom of THF. Two of these interactions are complementary, resulting in the formation of an eight-membered $\{Li_2O_4H_2\}$ ring (involving two molecules each of THF and water, as well as two lithium cations) between two $\{(H_2O)_2(THF)Li(BBB)\}$ units (Figure 8). This arrangement is unprecedented.

Indeed, mixed solvates of lithium cations in general are relatively rare, and the arrangement observed for $\{(H_2O)_2(THF)Li(BBB)\}_\infty$ is the first example of lithium bound to one THF and two water molecules. Related arrangements are limited to the following: (1) two examples where lithium is bound to one molecule each of water and THF;²³ (2) one example where lithium is bound to one water and two THF molecules;²⁴ and (3) one example where lithium is bound to two molecules of each.²⁵ Interestingly, none of these exhibit hydrogen bonding between solvent molecules. The last example, lithium tetraphenylborate, consists of a noncoordinating ion pair, $\{[B(C_6H_5)_4]^-[(H_2O)_2Li(THF)_2]^+\}$. Here the environment at lithium is essentially the same as in

Table 4. Calculated Bonding Characteristics for $Li-O_{BBB}$ and $Na-O_{BBB}$

bond	r [Å]	bond order	Mulliken population	Mulliken change
$Li-O^a$	1.922	0.101	2.1675 (Li)	0.8325
$Na-O^b$	1.922	0.083	10.187 (Na)	0.813
$Li-O^c$	1.856	0.091	2.1778 (Li)	0.8222
$Na-O^c$	2.127	0.084	10.301 (Na)	0.699

^a Using the experimentally observed crystal structure, without optimization. ^b Replacement of lithium by sodium in the experimentally observed crystal structure, without optimization. ^c After optimization.

$\{(H_2O)_2(THF)Li(BBB)\}_\infty$, except that one solvent molecule has been replaced by the coordinating $[BBB]^-$ anion. Unlike $\{(H_2O)_2(THF)Li(BBB)\}_\infty$, however, the hydrogen atoms of water in $\{[B(C_6H_5)_4]^-[(H_2O)_2Li(THF)_2]^+\}$ are oriented toward the faces of neighboring phenyl rings, and no $O-H\cdots O$ hydrogen bonding is observed at all.

The B–O bond distances in $\{(H_2O)_2(THF)Li(BBB)\}_\infty$ are somewhat shorter than the mean values for all structures containing $[BBB]^-$ anions [$\bar{r}(B-O) = 1.468(2), 1.48(1)$ Å, respectively].⁷ This is probably a consequence of electron withdrawal from the $[BBB]^-$ unit by (1) hydrogen bonding; and (2) the $\{(H_2O)_2Li(THF)\}^+$ unit. To probe whether this latter effect resulted in partial covalency of the $Li-O_{BBB}$ bond ($Li-O$ bonds are often considered to be essentially ionic^{8f}), we used the ab initio package GAMESS²⁶ to determine the location of electron density within the aggregate. Since the structure of the complex had already been determined by crystallographic methods, energy optimization was employed only to compare the optimized bonding characteristics with that of the experimentally observed structure. In addition, the $Li-O_{BBB}$ bond was compared with a $Na-O_{BBB}$ bond by replacing the lithium atom in $\{(H_2O)_2(THF)Li(BBB)\}$ with a sodium atom and repeating the optimization and calculations. These results are summarized in Table 4, and planar electron density cuts are shown in Figure 9. The calculated atomic charges suggest that both the $Li-O_{BBB}$ and $Na-O_{BBB}$ bonds are more ionic than covalent, and indeed, there does not appear to be a significant molecular orbital involving both atoms in either case (ca. 10% Mulliken bond order). However, comparison of the two structures suggests that the $Li-O_{BBB}$ bond is the more covalent of the two, the $Na-O_{BBB}$ bond being of lower bond order and thus higher ionic character. While the calculated $Na-O_{BBB}$ distance (2.127 Å) is somewhat shorter than the mean value observed experimentally [$\bar{r}(Na-O) = 2.43(14)$ Å], it well within the range of 1.755–3.288 Å previously observed for such contacts.⁵

The arrangement adopted by $\{[(H_2O)(MeCN)Li(BBB)]_2\}_\infty$ differs from that observed for $\{(H_2O)_2(THF)Li(BBB)\}_\infty$ as a consequence of the cosolvent employed. Whereas THF can act as a hydrogen-bond acceptor for water in the former, MeCN does not participate in hydrogen bonding in the latter. To achieve coordinative saturation of the anion and cation in $\{[(H_2O)(MeCN)Li(BBB)]_2\}_\infty$, each lithium cation is associated with two anions [$r(Li-O_{BBB}) = 1.977(4), 1.997(4)$ Å], dimerization occurring through an eight-membered $\{Li_2O_4B_2\}$ ring to form $\{(H_2O)(MeCN)Li(BBB)\}_2$ units. Consequently, the coordination environment around lithium includes

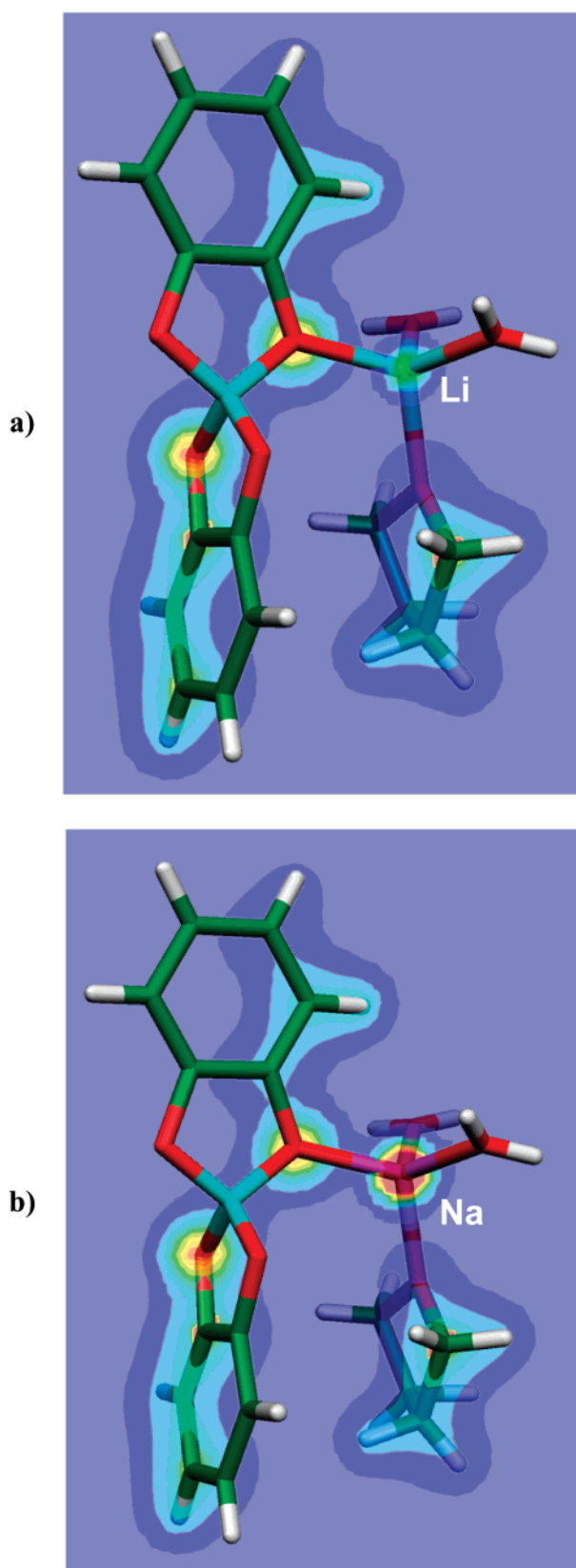


Figure 9. Electron density plots of the geometry-optimized structures of (a) $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}$; and (b) $\{(\text{H}_2\text{O})_2(\text{THF})\text{Na}(\text{BBB})\}$. Red coloration indicates areas of higher electron density, and blue of lower.

only single ethanenitrile [$r(\text{Li}-\text{N}) = 1.997(4) \text{ \AA}$] and water [$r(\text{Li}-\text{O}_{\text{water}}) = 1.882(4) \text{ \AA}$] molecules (Figure 10). Each water molecule is also hydrogen bonded to two

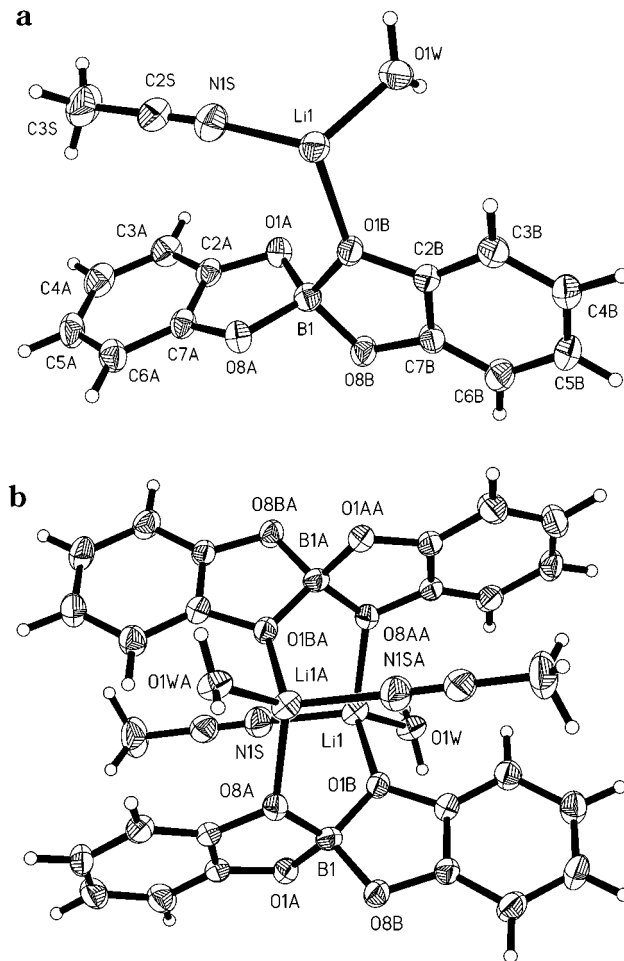


Figure 10. Thermal ellipsoid plot (50% probability) of (a) one monomer unit contained in the crystal structure of $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$, showing the atom labeling scheme; (b) a dimeric $\{(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})\}_2$ unit.

other $\{(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})\}_2$ units, resulting in the formation of discrete, two-dimensional “sheets” in the *ab* crystal plane (Figure 11); these sheets have essentially no empty space between them, and no channels or cavities are formed. Dimerization through Li–O bonds, rather than through hydrogen bonding as in $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$, apparently alleviates the need for a second water molecule in the structure and prevents the formation of a three-dimensional lattice. This unique arrangement is the first example of a mixed solvate of lithium involving ethanenitrile.

In cases where the availability of coordinating solvent is limited, oligomerization is often the only way to satisfy the various bonding requirements of organolithium compounds. The tradeoff between oligomerization and solvation (and the resultant diversity in molecular architecture) is a well-known phenomenon, and has been discussed in detail elsewhere.⁸ The common driving force for the topologies of all four structures discussed here is the achievement of coordinative saturation of the cation and anion via solvation (and, where necessary, oligomerization). In each case, lithium becomes four-coordinate through contacts with anions and solvent molecules, with the choice of solvent molecules being dictated by the remaining coordination requirements of the anions. When water is required, but only adventitious amounts are available, the remaining

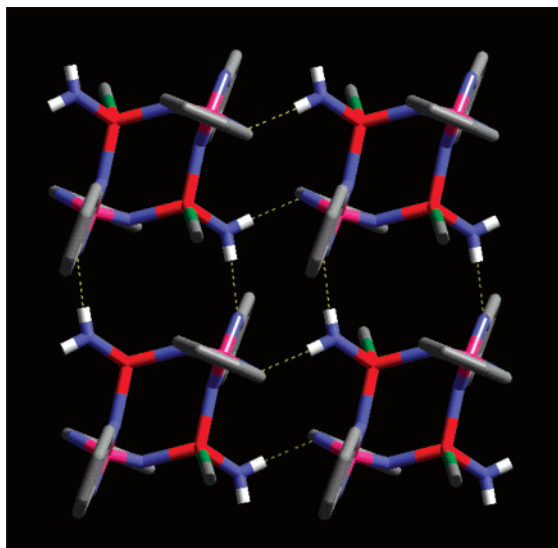


Figure 11. A portion of the of the hydrogen-bonded packing arrangement observed in the crystal structure of $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$, looking down the *c*-axis. All hydrogen atoms except those on water have been omitted for clarity. Color scheme as in Figure 2.

coordination sites at lithium are occupied by the bulk solvent. Thus, lithium salts of the bidentate, bridging $[\text{BSB}]^-$ anion are likely to be polymeric in the solid state in any situation where donor solvents are available to complete the coordination environment at lithium. Adventitious water is not incorporated because this polymeric arrangement satisfies both the cation and anion from a coordination perspective. In contrast, the tetradentate, bridging $[\text{BBB}]^-$ anion must to utilize both water molecules and oligomerization to saturate the bonding environments of the cation and anion. Regardless of the cosolvent present, this will result in extended networks (involving hydrogen bonding between hydrated lithium cations and anions) rather than single-strand polymers for this anion.

Finally, we note that while dissolution of $\text{Li}(\text{BBB})$ and $\text{Li}(\text{BSB})$ in water yields catechol and salicylic acid, respectively, crystals of $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$ and $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$ are stable in their mother liquors for periods in excess of two months. Apparently, when only adventitious water is available, the formation of a hydrogen-bonded lattice is thermodynamically preferable to decomposition of the $[\text{BBB}]^-$ anion through hydrolysis. This is likely due to both: (a) a paucity of water molecules, and (b) the shape and hydrogen-bond acceptor ability of the $[\text{BBB}]^-$ anion, which allows for unusual hydrogen bonding arrangements involving solvated lithium cations. The same is *not* true of the $[\text{BSB}]^-$ anion, which does not form such lattices even when wet solvents are used for crystallization.

While both $\text{Li}(\text{BSB})$ and $\text{Li}(\text{BBB})$, as well as a number of derivatives and related compounds, have been examined from an electrochemical perspective previously,^{3,4} the results presented here are the first crystallographic studies of lithium organoborates. Investigation of these and similar compounds is ongoing in our laboratories, and we anticipate that gaining a fundamental understanding of such materials will facilitate their eventual use in lithium batteries and elsewhere.

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Supporting Information Available: X-ray diffraction data (CIF) including tables of atomic coordinates, thermal parameters, crystallographic data, and bond distances and angles for $\{(\text{MeCN})_2\text{Li}(\text{BSB})\}_\infty$, $\{(\text{THF})_2\text{Li}(\text{BSB})\}_\infty$, $\{(\text{H}_2\text{O})_2(\text{THF})\text{Li}(\text{BBB})\}_\infty$, and $\{[(\text{H}_2\text{O})(\text{MeCN})\text{Li}(\text{BBB})]_2\}_\infty$. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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